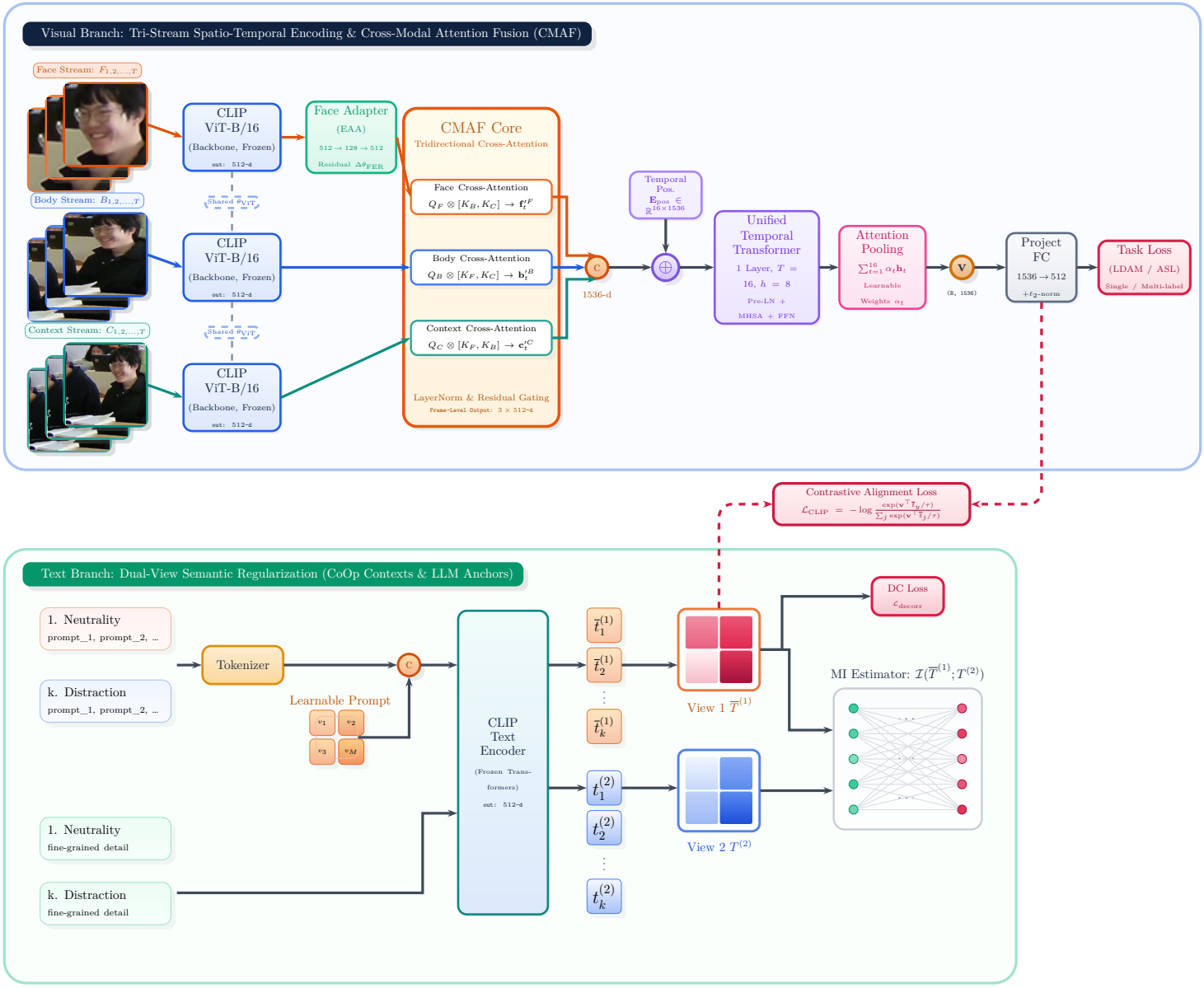

Visual Branch: Tri-Stream Spatio-Temporal Encoding & Cross-Modal Attention Fusion (CMAF)
Face Stream: $F_{1,2,\ldots,T}$
Body Stream: $B_{1,2,\ldots,T}$
Context Stream: $C_{1,2,\ldots,T}$
CLIP ViT-B/16 (Backbone, Frozen) out: 512-d
Shared $\theta_{\text{ViT}}$
Face Adapter (EAA) $512 \to 128 \to 512$ Residual $\Delta\theta_{\text{FER}}$
CMAF Core Tridirectional Cross-Attention
Face Cross-Attention $Q_F \otimes [K_B, K_C] \to \mathbf{f}_t'^F$
Body Cross-Attention $Q_B \otimes [K_F, K_C] \to \mathbf{b}_t'^B$
Context Cross-Attention $Q_C \otimes [K_F, K_B] \to \mathbf{c}_t'^C$
LayerNorm & Residual Gating Frame-Level Output: $3 \times 512$-d
1536-d
Temporal Pos. $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{16\times1536}$
Unified Temporal Transformer 1 Layer, $T = 16$, $h = 8$ Pre-LN + MHSA + FFN
Attention Pooling $\sum_{t=1}^{16} \alpha_t \mathbf{h}_t$ Learnable Weights $\alpha_t$
(B, 1536)
Project FC $1536 \to 512$ $+\ell_2$-norm
Task Loss (LDAM / ASL) Single / Multi-label
Contrastive Alignment Loss $\mathcal{L}_{\text{CLIP}} = -\log \frac{\exp(\mathbf{v}^\top \mathbf{t}_y/\tau)}{\sum_j \exp(\mathbf{v}^\top \mathbf{t}_j/\tau)}$
Text Branch: Dual-View Semantic Regularization (CoOp Contexts & LLM Anchors)
1. Neutrality prompt_1, prompt_2, ...
k. Distraction prompt_1, prompt_2, ...
Tokenizer
Learnable Prompt
$v_1$ $v_2$ $v_3$ $v_M$
CLIP Text Encoder (Frozen Transformers) out: 512-d
$\bar{t}_1^{(1)}$ $\bar{t}_2^{(1)}$ $\bar{t}_k^{(1)}$
View 1 $\overline{T}^{(1)}$
DC Loss $\mathcal{L}_{\text{decorr}}$
MI Estimator: $\mathcal{I}(\overline{T}^{(1)}; T^{(2)})$
$t_1^{(2)}$ $t_2^{(2)}$ $t_k^{(2)}$
View 2 $T^{(2)}$
1. Neutrality fine-grained detail
k. Distraction fine-grained detail
Feature Concatenation
Element-wise Addition
Face Stream
Body Stream
Context Stream
Contrastive Alignment